

Sharp Analysis of Gaussian Rounding for Boolean Max k -CSP

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In this note, we show that the approximation algorithm for Boolean Max k -CSP presented in [7] yields a $(1 - o_k(1))k/2^k$ approximation, as conjectured in [8]. This improves the previous guarantee of $(0.626612 - o_k(1))k/2^k$ from [7] and asymptotically matches the known hardness results; see [8, 1, 5, 3]. The result is a short corollary of the Gaussian stochastic domination theorem of Mulgund [9].

1 Introduction

For Boolean Max k -CSP, Hast [6] gave an $\Omega(k/(2^k \log k))$ approximation, and Charikar, Makarychev, and Makarychev [4] obtained a $ck/2^k$ approximation for an absolute constant $c > 0.44$. In [7] we improved the constant to $0.626612 - o_k(1)$ and conjectured that the same approach gives a $(1 - o_k(1))k/2^k$ approximation; see also the survey [8].

The approximation guarantee is asymptotically optimal. Austrin and Mossel [1] showed, assuming the Unique Games Conjecture, hardness of approximation within $(k + O(k^{0.525}))/2^k + \varepsilon$. De and Mossel [5] subsequently improved this to $(k + 1)/2^k + \varepsilon$ for odd k and $(k + 2)/2^k + \varepsilon$ for even k . Thus, under the Unique Games Conjecture, the optimal approximation factor is $(1 + o_k(1))k/2^k$. Chan [3] obtained unconditional hardness from pairwise-independent subgroups; in particular, for $k = 2^r - 1$, his result gives hardness within $(k + 1)/2^k + \varepsilon$. Hence the constant in our approximation guarantee is optimal under $P \neq NP$ for an infinite sequence of values of k .

Overview. We use the same SDP relaxation, notation, and Gaussian rounding as in [7]. The only new ingredient is Mulgund's stochastic domination theorem [9], which we use as a black box. If C is a clause and $\|z_C\|^2 = \beta^2/k$, the theorem gives the key estimate

$$\Pr(C \text{ is satisfied}) \geq \int_{\mathbb{R}} F(cx)^k d\gamma(x), \quad c^2 = \frac{\beta^2 - 1}{k - \beta^2}.$$

At $\|z_C\|^2 = 1/k$ the right-hand side is exactly 2^{-k} , while slightly above this value it grows sufficiently fast. Combining this rounding with a random assignment gives the claimed $(1 - o_k(1))k/2^k$ approximation. Thus the result is essentially a corollary of [9].

Related work: Concurrently and independently, Bakshi [2] also settled the conjecture for Boolean Max k -CSP.

2 The algorithm

An instance of Boolean Max k -CSP consists of Boolean variables $x_u \in \{1, 2\}$ and constraints, each depending on at most k variables; the goal is to find an assignment that satisfies the maximum number of

constraints. Following Trevisan [10], as in [7], we may replace a predicate $P(x_{v_1}, \dots, x_{v_r})$ by the clauses

$$\{(x_{v_1} = i_1) \wedge \dots \wedge (x_{v_r} = i_r) : P(i_1, \dots, i_r) = 1\}.$$

For every assignment, exactly one of these clauses is satisfied if P is satisfied, and none is satisfied otherwise. Thus this transformation preserves the value of every assignment. We thus assume below that every constraint is a conjunction of literals. We use the same SDP relaxation and notation as in [7]. For every variable x_u and $i \in \{1, 2\}$ the SDP has a vector u_i , and for every clause C it has a vector z_C :

$$\begin{aligned} & \text{maximize: } \sum_{C \in \mathcal{C}} \|z_C\|^2 \\ & \text{subject to} \\ & \|u_1\|^2 + \|u_2\|^2 = 1 \\ & \langle u_1, u_2 \rangle = 0, \\ & \langle u_i, z_C \rangle = \|z_C\|^2 \quad \text{if } (u, i) \in C, \\ & \langle u_j, z_C \rangle = 0 \quad \text{if } (u, i) \in C, j \neq i. \end{aligned}$$

The intended meaning of these vectors is the following. In an integral solution, for a fixed unit vector $\mathbf{e}$, we set $u_i = \mathbf{e}$ if $x_u = i$ and $u_i = 0$ otherwise; similarly, $z_C = \mathbf{e}$ if C is satisfied and $z_C = 0$ otherwise. Thus $\|u_i\|^2$ is a relaxation of the indicator that $x_u = i$, while $\|z_C\|^2$ is a relaxation of the indicator that C is satisfied. The constraints above are satisfied by every integral assignment, therefore, $SDP \geq OPT$. We may also assume that every clause has length exactly k , by padding shorter clauses with extra variables.

Rounding procedure The SDP rounding is the same Gaussian rounding as in [7]: choose a standard Gaussian vector g and set

$$x_u = \arg \max_{i \in \{1, 2\}} \langle u_i, g \rangle.$$

The final algorithm uses this rounding with probability $1/k$ and a uniformly random assignment with probability $1 - 1/k$. Write

$$F(t) = \Pr(g \leq t),$$

where $g \sim \mathcal{N}(0, 1)$, and denote standard Gaussian measure on $\mathbb{R}$ by γ . We use the following theorem of Mulgund [9].

Theorem 1 (Mulgund). *Let R be a $k \times k$ correlation matrix and let J be the all-ones matrix. If*

$$R - \frac{1}{k}J \succeq 0$$

and $X \sim \mathcal{N}(0, R)$, then, for every $t \in \mathbb{R}$,

$$\Pr(X_u \leq t \text{ for every } u) \geq F(t)^k.$$

3 Analysis of one clause

Fix a clause C . As in [7], we may assume without loss of generality that $C = \{(u, 1) : u \in \text{supp}(C)\}$, that is, the clause is of the form $\bigwedge_{u \in \text{supp}(C)} (x_u = 1)$. Crucially, the clause C is satisfied if and only if

$$\langle v_u, g \rangle \geq 0 \quad \text{for every } u \in \text{supp}(C)$$

where $g \sim \mathcal{N}(0, I)$ is the random Gaussian vector from the rounding scheme. Our goal now is to lower bound the probability of this event.

Define $s = \|z_C\|^2$ and, for $u \in \text{supp}(C)$,

$$v_u = u_1 - u_2 \quad \text{and} \quad w_u = v_u - z_C.$$

By the SDP constraints, we have $\|v_u\|^2 = \|u_1\|^2 + \|u_2\|^2 = 1$, $\langle v_u, z_C \rangle = s$, $w_u \perp z_C$, and $\|w_u\|^2 = 1 - s$. Let Σ be the Gram matrix of the vectors v_u , $u \in \text{supp}(C)$. Since all vectors v_u share the same component z_C , namely $v_u = z_C + w_u$ for $u \in \text{supp}(C)$, where $w_u \perp z_C$, we can write

$$\langle v_a, v_b \rangle = s + \langle w_a, w_b \rangle,$$

or, for $s < 1$, in the matrix form

$$\Sigma = sJ + (1 - s)G, \tag{1}$$

where G is defined as

$$G_{uv} = \frac{\langle w_u, w_v \rangle}{1 - s}.$$

Note that the correlation matrix of random variables $\xi_u = \langle v_u, g \rangle$ is exactly Σ . We would like to apply Theorem 1 now. However, applying it directly to ξ_u will not yield the desired bound. Let $\beta = \sqrt{sk}$ and assume that $\beta \geq 1$. We add a part of the common component sJ to G , so that after normalization the resulting matrix dominates J/k , and keep the rest separate:

$$\Sigma = (1 - s) \left(G + \frac{1}{k-1} J \right) + \frac{ks-1}{k-1} J.$$

Define

$$R = \frac{k-1}{k} \left(G + \frac{1}{k-1} J \right) = \frac{k-1}{k} G + \frac{1}{k} J.$$

Then R is a correlation matrix and

$$R - \frac{1}{k} J = \frac{k-1}{k} G \succeq 0.$$

Let $\theta = \frac{k-\beta^2}{k-1}$, using $s = \beta^2/k$, we get

$$\Sigma = \theta R + (1 - \theta)J.$$

Consequently, if $X \sim \mathcal{N}(0, R)$ and $Z \sim \mathcal{N}(0, 1)$ are independent, then

$$(\xi_u)_{u \in \text{supp}(C)} \stackrel{d}{=} (\sqrt{\theta}X_u + \sqrt{1-\theta}Z)_{u \in \text{supp}(C)}.$$

Applying Theorem 1 after conditioning on Z gives the key estimate

$$\Pr(C \text{ is satisfied}) = \Pr \left(-X_u \leq \sqrt{\frac{1-\theta}{\theta}} Z \text{ for all } u \in \text{supp}(C) \right) \geq \int_{\mathbb{R}} F(cx)^k d\gamma(x), \tag{2}$$

$$\text{where } c^2 = \frac{1-\theta}{\theta} = \frac{\beta^2-1}{k-\beta^2}. \tag{3}$$

Claim 1. *Let*

$$\beta_k^2 = 1 + \frac{4\pi \log k}{k}.$$

For all sufficiently large k , if

$$\|z_C\|^2 \geq \frac{\beta_k^2}{k},$$

then the SDP rounding satisfies C with probability at least $k^2/2^k$.

Proof. It suffices to estimate (2) at $\beta = \beta_k$. Indeed, c is increasing in β , and the integral is increasing in $c \geq 0$: pairing x and $-x$, the integrand becomes $p^k + (1-p)^k$, where $p = F(cx) \geq 1/2$, and this is increasing in p . Let c_k be the value of c corresponding to $\beta = \beta_k$. Then

$$c_k^2 = (4\pi + o(1)) \frac{\log k}{k^2}.$$

Put $a = \sqrt{2/\pi}$ and $x_k = akc_k$. Since $\log(2F(t)) = at + O(t^2)$ (when $t \rightarrow 0$) and $c_k x_k = akc_k^2 = o(1)$, we have

$$\begin{aligned} \log \left(2^k \int_{\mathbb{R}} F(c_k x)^k d\gamma(x) \right) &\geq k \log(2F(c_k x_k)) + \log \gamma([x_k, x_k + 1]) \geq a^2 k^2 c_k^2 - \frac{x_k^2}{2} + o(k^2 c_k^2) \\ &= \left(\frac{1}{\pi} + o(1) \right) k^2 c_k^2 = (4 + o(1)) \log k. \end{aligned}$$

Hence

$$\int_{\mathbb{R}} F(c_k x)^k d\gamma(x) \geq \frac{k^{4+o(1)}}{2^k} \geq \frac{k^2}{2^k}.$$

The case $\|z_C\| = 1$ is immediate. $\square$

4 Approximation guarantee

Theorem 2. *There is a randomized polynomial-time approximation algorithm for Boolean Max k -CSP with approximation guarantee*

$$\left(1 - O \left(\frac{\log k}{k} \right) \right) \frac{k}{2^k}.$$

In particular, the approximation guarantee is $(1 - o_k(1))k/2^k$.

Proof. Let

$$\alpha_k = \frac{1 - 1/k}{\beta_k^2} = 1 - O \left(\frac{\log k}{k} \right).$$

Consider a clause C and let $s = \|z_C\|^2$. If $s < \beta_k^2/k$, the random assignment satisfies C with probability at least

$$\frac{1 - 1/k}{2^k} \geq \alpha_k \frac{ks}{2^k}.$$

If $s \geq \beta_k^2/k$, Claim 1 and the rounded assignment satisfies C with probability at least

$$\frac{1}{k} \cdot \frac{k^2}{2^k} = \frac{k}{2^k} \geq \frac{ks}{2^k} \geq \alpha_k \frac{ks}{2^k}.$$

Thus every clause C is satisfied with probability at least

$$\alpha_k \frac{k}{2^k} \|z_C\|^2.$$

Summing over all clauses,

$$\mathbb{E}[\text{number of satisfied clauses}] \geq \alpha_k \frac{k}{2^k} \sum_{C \in \mathcal{C}} \|z_C\|^2 = \alpha_k \frac{k}{2^k} SDP \geq \alpha_k \frac{k}{2^k} OPT.$$

$\square$

References

- [1] Per Austrin and Elchanan Mossel. Approximation resistant predicates from pairwise independence. *Comput. Complexity*, 18(2):249–271, 2009. Preliminary version in CCC’08.
- [2] Ainesh Bakshi. On the Approximability of Boolean Max- k -CSP. *CoRR*, abs/2608.05331, 2026. arXiv:2608.05331.
- [3] Siu On Chan. Approximation resistance from pairwise-independent subgroups. *Journal of the ACM (JACM)*, 63(3):1–32, 2016.
- [4] Moses Charikar, Konstantin Makarychev, and Yury Makarychev. Near-optimal algorithms for maximum constraint satisfaction problems. *ACM Trans. Algorithms*, 5(3), 2009. Preliminary version in SODA’07.
- [5] Anindya De and Elchanan Mossel. Explicit optimal hardness via gaussian stability results. *ACM Transactions on Computation Theory (TOCT)*, 5(4):1–26, 2013.
- [6] Gustav Hast. Approximating Max k CSP — outperforming a random assignment with almost a linear factor. In *Proc. Intl. Colloquium on Automata, Languages, and Programming (ICALP)*, pages 956–968. Springer-Verlag, 2005.
- [7] Konstantin Makarychev and Yury Makarychev. Approximation Algorithm for Non-Boolean Max- k -CSP. *Theory of Computing*, 10(13):341–358, 2014. Preliminary version in APPROX 2012, pp. 254–265.
- [8] Konstantin Makarychev and Yury Makarychev. Approximation Algorithms for CSPs. In Andrei Krokhin and Stanislav Zivny, editors, *The Constraint Satisfaction Problem: Complexity and Approximability*, volume 7 of *Dagstuhl Follow-Ups*, pages 287–325. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2017.
- [9] Abhijeet Mulgund. Stochastic Domination of Gaussian Maxima: A Resolution to the Weak Simplex Conjecture. *CoRR*, abs/2607.14087, 2026. arXiv:2607.14087.
- [10] Luca Trevisan. Parallel approximation algorithms by positive linear programming. *Algorithmica*, 21(1):72–88, 1998.